Part 7: Landing on Planet 1

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We landed a 90 kg lander on planet 1, 0.7 km west of the site chosen in Part 6, at 1.98 m/s (limit 3 m/s), in full daylight with the star 86° above the horizon.

- **Analysis.** In air that co-rotates with the planet, a parachute of at least 11.44 m^2 brings the lander below 3 m/s. Without one it would hit the ground at 18.5 m/s.
- **Simulator.** Our own descent simulator includes gravity, drag in the rotating air, the lander release and the parachute. It agrees with `ast2000tools` [1] to 1.03×10^3 m over a whole test descent.
- **The landing.** The lander is released from the low orbit with 200 m/s against its motion and brakes from 4.3 km/s in the upper atmosphere. A 26 m^2 parachute opens 3 km above the ground. The release time comes from a root search with our simulator, and the real flight landed within 0.74 km of the predicted point.
- **Margins.** The peak drag per area was 861 times below the burn-up limit. The parachute's peak load was 40.7 kN, against a 250 kN limit.

I. RESULTS

A. Air resistance and terminal velocity

We use the drag model of the assignment [2]. The atmosphere co-rotates with the planet, so the air at $\mathbf{r}$ moves with $\mathbf{w} = \omega\hat{\mathbf{z}} \times \mathbf{r}$, which is 456 m/s at the equator. Drag acts on the velocity relative to the air, $\mathbf{v}_{\mathrm{drag}} = \mathbf{v} - \mathbf{w}$:

$$\mathbf{F}_d = -\tfrac{1}{2}\rho(h) C_d A \,|\mathbf{v}_{\mathrm{drag}}|\, \mathbf{v}_{\mathrm{drag}}, \qquad C_d = 1. \tag{1}$$

Two facts follow from this.

- The drag always opposes the relative velocity, and gravity is radial. Nothing sustains a tangential relative velocity, so $v_\theta \to 0$ relative to the air.
- Drag grows as v^2 until it balances gravity, so v_r settles at the terminal velocity

$$v_t = \sqrt{\frac{2GMm}{\rho A R^2}} \quad \Longrightarrow \quad A = \frac{2GMm}{\rho_0 R^2 v^2}. \tag{2}$$

For our lander:

- Alone it lands at $v_t = 18.5$ m/s, which would be a crash.
- A total area of 11.44 m^2 gives exactly 3 m/s (Fig. 1).
- Opening at terminal speed, the parachute's drag is $mg\, A_{\mathrm{tot}}/A_{\mathrm{lander}}$. The largest chute that survives opening is therefore 160 m^2.
- Thrusters alone would need $F_L = \frac{1}{2}\rho_0 A(v_t^2 - v_{\mathrm{safe}}^2) =$ 456 N, almost the lander's weight (Fig. 2).

We use a 26 m^2 parachute, sized for 2 m/s at the ground, and no thrusters.

B. The simulator

Our simulator (Fig. 3) integrates gravity, drag in the co-rotating air (density from the Part 6 model), the lander release, the parachute and the thrusters. It uses an adaptive eighth-order Runge–Kutta solver (DOP853 [3]), and the ground is detected as an event. The adaptive steps handle the violent moments automatically: entry, the parachute opening, decelerations of $\sim 100\,g$. On a test descent (release with 150 m/s, 26 m^2 parachute at 10 km) it agrees with `ast2000tools` to within 20 m before entry and 1.03×10^3 m after (Fig. 4). The remaining difference matches the 1–2% between our density model and the API's (Part 6).

C. The landing

The lander's landing longitude depends on when it is released. Once braked it drifts with the air, so its planet-fixed position hardly changes afterwards. Scanning the release time over two orbits gives 2 releases that land on site 3 (Fig. 5). We take the one in sunlight: release at $t = 5.762$ h.

In the real flight, the spacecraft oriented itself at that time and released the lander with 200 m/s against its measured velocity (Fig. 6).

- **Braking:** it slowed from 4.3 km/s high in the atmosphere, at a peak drag per area of 1.16×10^4 Pa, 861 times below the burn-up limit.
- **Parachute:** it opened 47.3 min after the release, at 3 km, with a peak load of 40.7 kN.
- **Touchdown:** 72.0 min after the release, at $t =$ 6.962 h.

`ast2000tools` reported a successful landing at 1.98 m/s, against our prediction of 1.97 m/s (Fig. 7). The landing

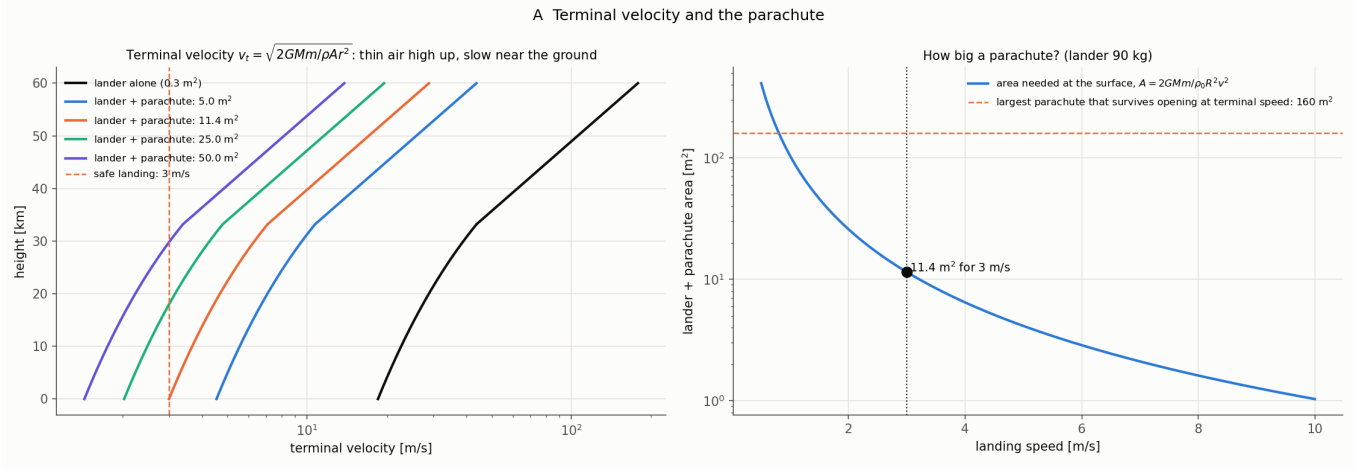


FIG. 1. Terminal velocity against height for the lander with parachutes of several sizes, and the area needed for a given landing speed.

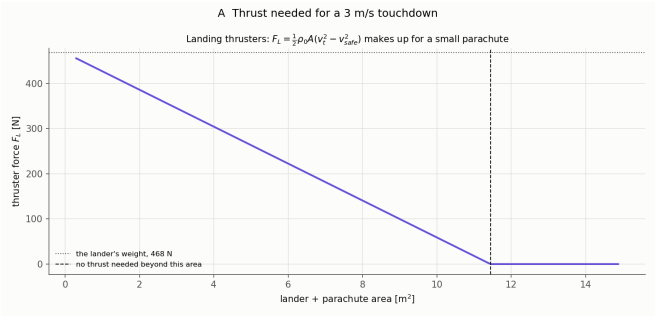


FIG. 2. Thruster force needed for a 3 m/s touchdown against the total area.

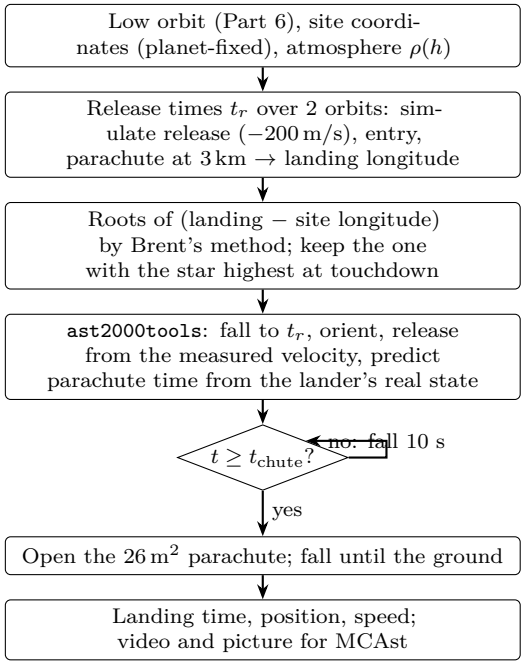


FIG. 3. Flowchart of the landing code.

point is 0.7 km west of site 3 along the equator (Fig. 8), with the star 86° high. The landing video (`landing.xml`) and a picture from the surface (`landed.xml`) are in MCAst.

II. DISCUSSION AND CONCLUSIONS

Physics alone made the landing robust: no trial and error was needed.

- The lander's small area per mass makes it brake high, where the air is thin. The drag per area stays far below the burn-up limit even for a direct entry.
- Opening the parachute only once the lander falls at its own terminal speed caps the opening load at $mg\,A_{\text{tot}}/A_{\text{lander}}$, independent of height.
- Opening it low, at 3 km, keeps the slow final descent short. Higher openings make the touchdown time, and so the lighting, harder to predict.

The landing accuracy comes from two things:

- a simulator that matches the API to a fraction of a percent of the trip;
- re-planning from the measured state at release.

The remaining 0.74 km follows from small differences between our atmosphere model and the API's in the upper atmosphere, where the lander brakes. This is the same order as the difference measured in Part 6. A course correction during the parachute phase is not possible. A smaller miss would need a better density model at 80–120 km, or iterating the release time against the API.

The lander is on planet 1, on the day side, 0.7 km from the chosen site, after touching down at 1.98 m/s.

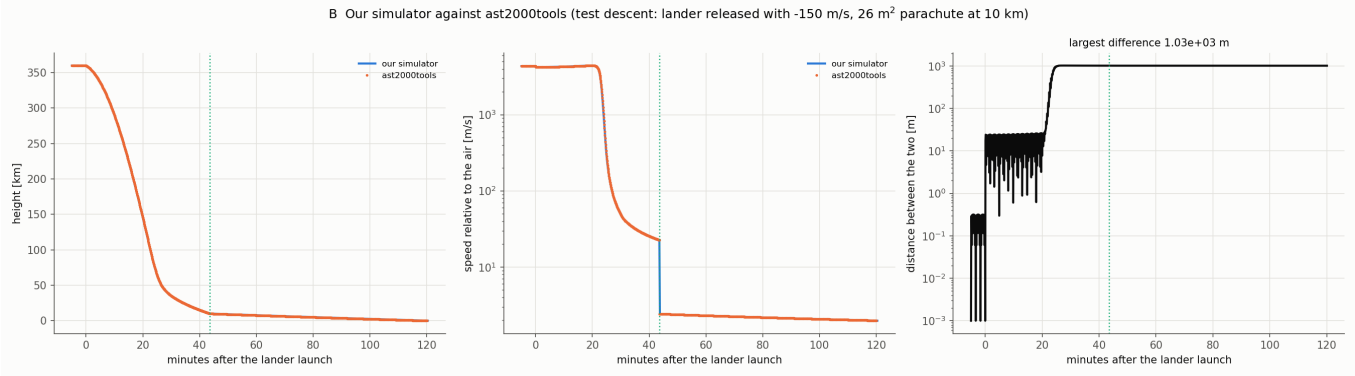


FIG. 4. Our simulator against `ast2000tools` on a test descent: height, speed relative to the air, and the distance between the two.

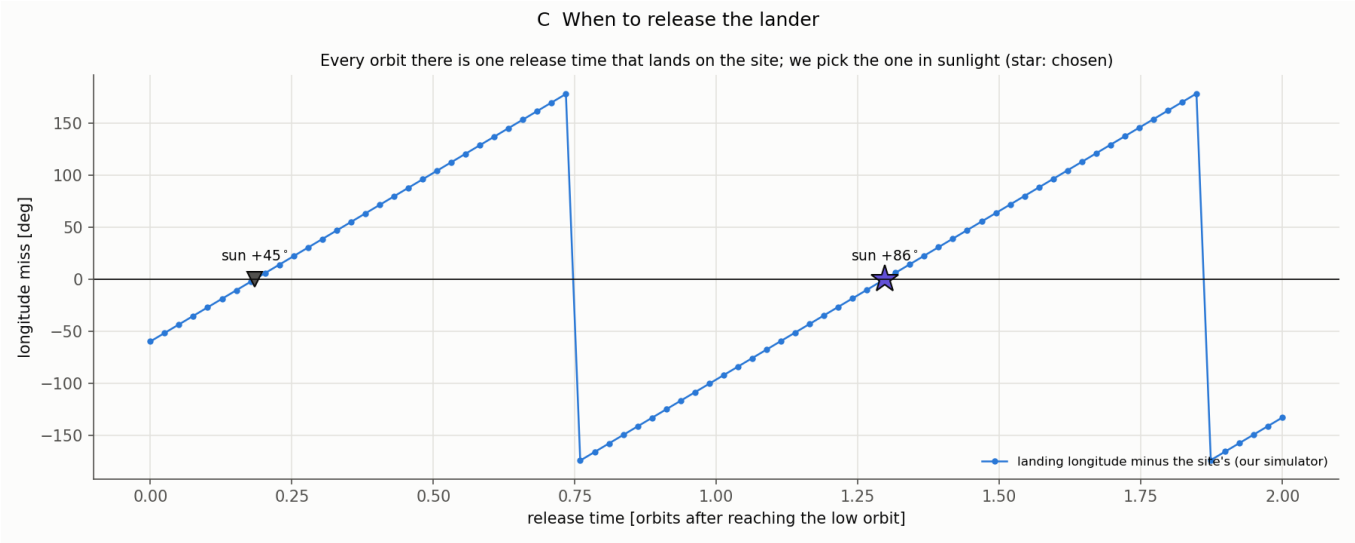


FIG. 5. Landing longitude miss against release time; the chosen release is the star.

[1] F. K. Hansen and L. F. Engedal, ast2000tools: Python package for the AST2000 project, https://frodekh.github.io/ast2000tools/html/index.html (2023).

[2] AST2000 teaching team, AST2000 spacecraft project, parts 1–9, Institute of Theoretical Astrophysics, University of Oslo (2023).

[3] J. R. Dormand and P. J. Prince, A family of embedded Runge-Kutta formulae, Journal of Computational and Applied Mathematics **6**, 19 (1980).

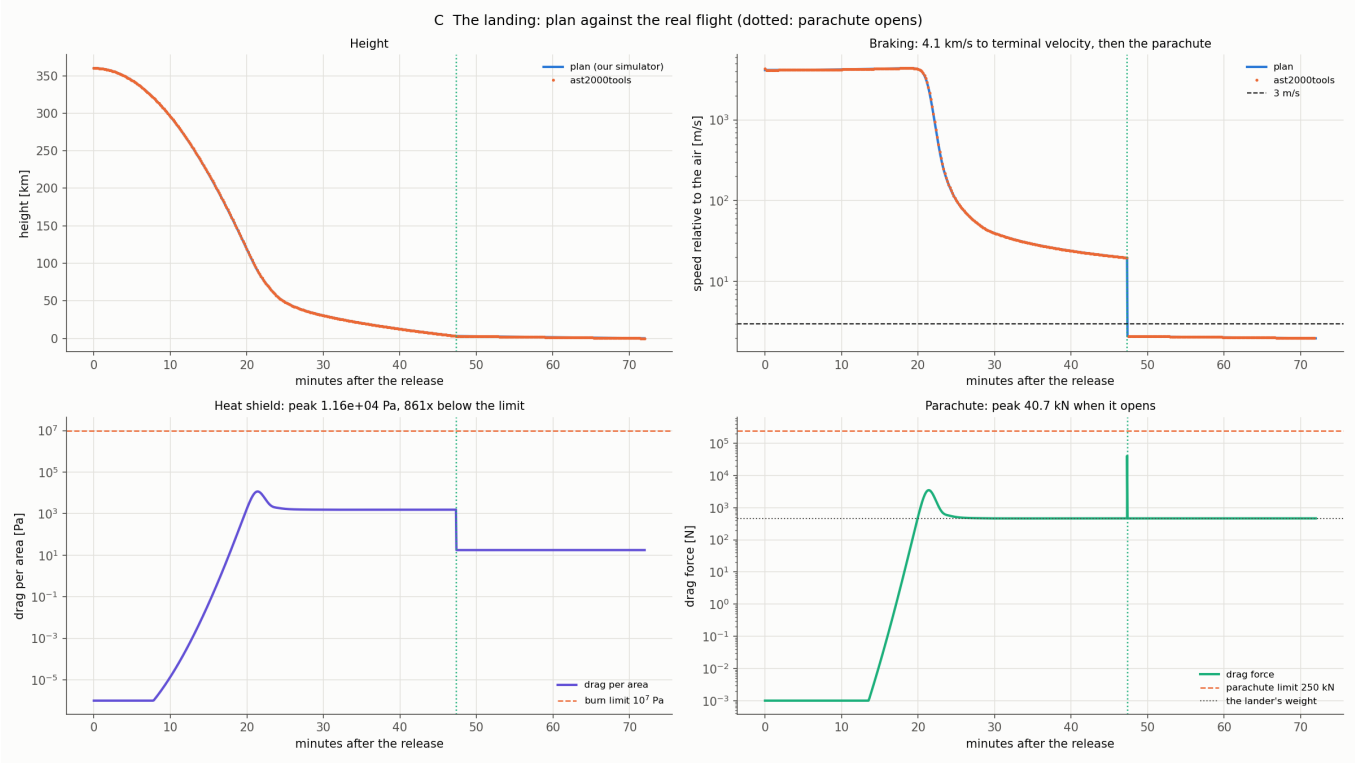


FIG. 6. The landing: planned against flown height and speed, drag per area against the burn-up limit, and drag force against the parachute limit.

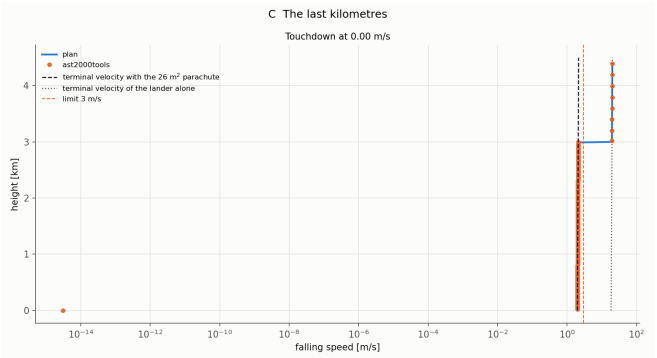


FIG. 7. The last kilometres: falling speed against height, with the terminal velocity curves.

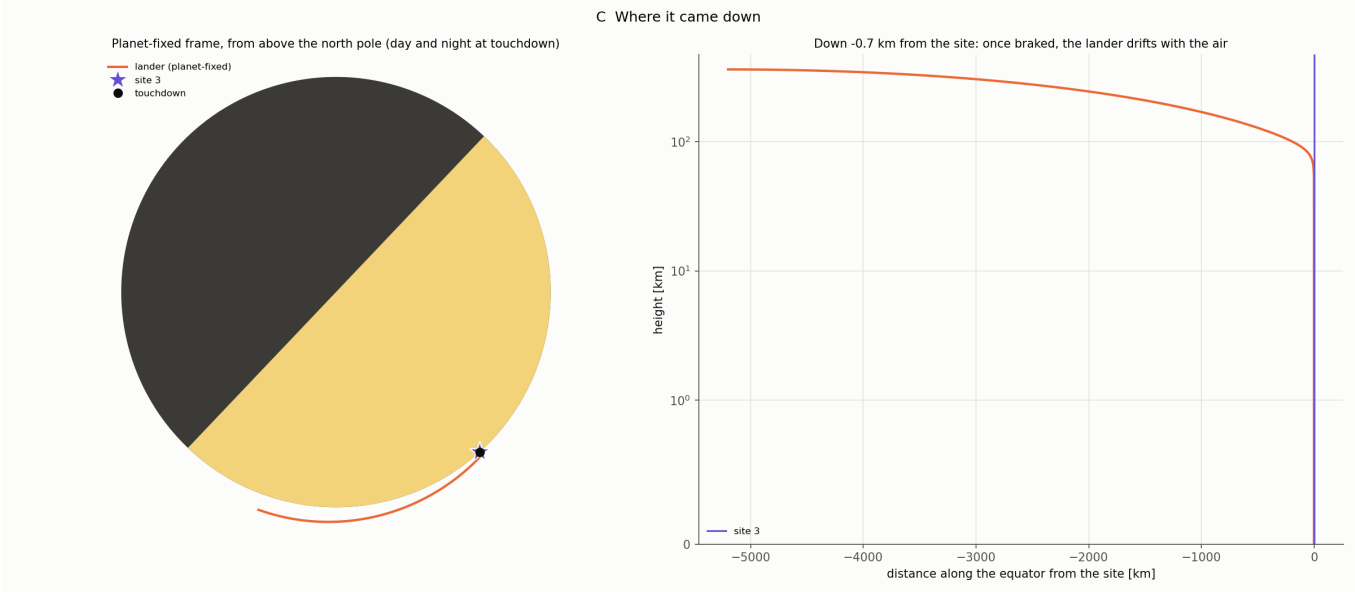


FIG. 8. Where it came down. Planet-fixed view from above the north pole, with day and night at touchdown, and height against distance from the site.